

AI/ML Detection of Malware C2 & DGA-Based DNS

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Project Proposal Report

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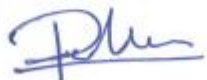
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DECLARATION

I hereby declare that this is my own work, and that this proposal has not incorporated, without proper acknowledgment, any material previously submitted for a degree or diploma at any other university or institution of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by any other person, except where explicit acknowledgment is made in the text.

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ABSTRACT

Attackers increasingly rely on Domain Generation Algorithms (DGAs) to create resilient Command and Control (C2) channels, rendering traditional defenses ineffective. Existing solutions often focus on isolated aspects of this problem or lack the interpretability required for effective security operations. This research proposes the design and development of a synergistic, multi-component framework to address these challenges comprehensively. The proposed system integrates three distinct components: a cascaded DGA detector combining a "Glass Box" rule engine and a "Black Box" deep learning model; a C2 detection engine using graph-based and flow-based behavioral analysis; and an advanced module for identifying encrypted C2 communication and integrating with Threat Intelligence platforms.

The core novelty lies in the synergistic architecture, particularly the use of the real-time DGA detector's output to dynamically "seed" the C2 graph analysis, creating a proactive framework capable of discovering novel threats. Further novelty is found in the framework's operational focus, demonstrated by its direct integration with a Wazuh SIEM to provide prioritized, actionable alerts. The anticipated outcome is a robust, interpretable, and efficient system that provides high-confidence, context-rich alerts for modern DNS-based threats.

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1. INTRODUCTION

In today's cybersecurity landscape, attackers use Command and Control (C2) servers to manage malware and botnets, posing a significant threat to the internet [1]. To avoid detection, modern malware employs Domain Generation Algorithms (DGAs) to generate thousands of pseudo-random domain names daily as rendezvous points with C2 servers [2]. Since these domains are constantly changing, traditional security methods based on static blocklists are no longer effective.

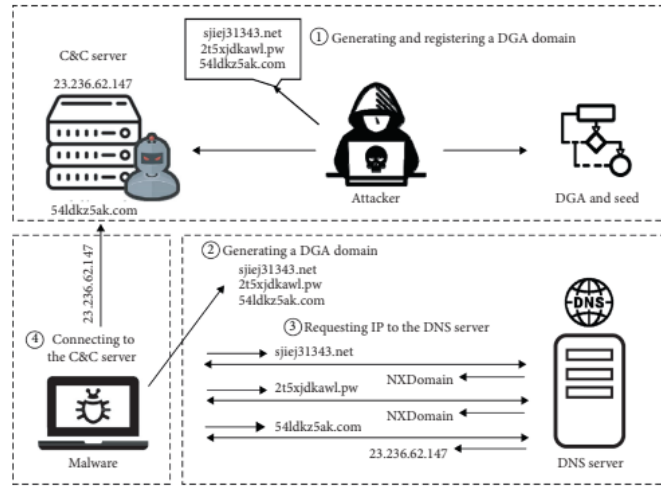


Figure 1: Attacker's procedure for using a DGA

The dynamic and evasive nature of DGA-based threats poses a significant challenge, mainly with two core components. The first is the accurate identification of individual malicious DGA domains from the massive volume of legitimate DNS traffic. The second is the ability to look beyond these single points of detection to uncover the entire hidden C2 communication infrastructure that attackers use to control their malware.

To tackle this, the current state of the art has moved towards machine learning and deep learning. Many early machine learning approaches relied on manually engineering lexical features from the domain string. However, this feature engineering is a tedious and time-consuming task that requires extensive domain knowledge [1].

More recently, deep learning models, particularly hybrid architectures like the CNN+BiLSTM, have demonstrated state-of-the-art performance by automatically learning the complex sequential patterns of DGA domains directly from the character sequence [2].

Some other researchers have used behavioral analysis to detect malicious activity in DNS traffic. Graph-based methods like GMAD track the sequence of DNS queries to reveal groups of domains that are often contacted together, which may indicate coordinated C2 infrastructure. Flow-based methods analyze traffic patterns, such as low DNS success rates, to detect bots attempting to resolve many invalid domains [3].

These approaches also have limitations. Many solutions focus on only one part of the problem, and the complexity of deep learning often results in "black box" models that lack the readability required for effective incident response. Critically, most of the C2 communication methods are rendered blind by the increasing use of encrypted protocols, a problem explicitly noted as an area for future work in several studies [4].

This research addresses those gaps by introducing a full detection framework that combines several specialized components. It uses a two-stage DGA detection pipeline: a fast, rule-based "Glass Box" engine for clear decisions, followed by a deep learning "Black Box" model for more complex cases. The output from this lexical analysis feeds into the C2 detection engine, which uses behavioral data for deeper analysis. Together, they provide stronger context and accuracy than using either approach alone. The system also includes modules for detecting encrypted C2 activity and integrates directly with the Wazuh SIEM for real-time, actionable alerts. Unlike many academic systems that stop at detection, this framework is built for operational use. Alerts are integrated into Wazuh SIEM, where they are normalized, scored, mapped to MITRE tactics, and visualized in real time. This integration is a core part of the novelty, turning raw detection into a practical, end-to-end security solution.

2. BACKGROUND & LITERATURE SURVEY

Detecting malicious DNS activity is a well-established area in cybersecurity, shaped by the ongoing evolution of attacker techniques. To support the proposed work, this

section reviews the core concepts and the current state of the art in detecting DGA-based domains and C2 communication.

2.1.Attacker Evasion Techniques

Botnets require a communication channel to a Command and Control (C2) server to receive instructions. To prevent these channels from being easily blocked, attackers employ sophisticated evasion techniques [1].

The most well-known technique is the use of Domain Generation Algorithms (DGAs). A DGA is an algorithm embedded within malware that generates a large number of pseudo-random domain names, only one of which the attacker needs to register to activate the C2 channel. Attackers also use the fast-flux technique [4], which involves rapidly changing the IP addresses associated with a single domain name, often using short Time-to-Live (TTL) values, making it difficult for defenders to block the C2 server's IP address. Advanced variants, known as dictionary DGAs, use real words to create domains that look more legitimate and are significantly harder for many detection models to identify [1].

2.2.Detection Approach 1: Feature-Based and Lexical Analysis

Early machine learning methods improved on static blocklists by focusing on the domain name itself. These models used feature engineering, where experts manually defined measurable properties, like domain length, randomness (entropy), vowel-to-consonant ratio, or unusual character patterns, to help the system tell apart normal domains and DGA-generated ones.

Key features explored in the literature include:

- **Lexical and Statistical Features:** This involves calculating measurable properties of a domain name, such as its length, character randomness (Shannon entropy), and the proportion of numbers to letters [4].
- **N-gram Analysis:** This technique breaks a domain name into short character sequences called n-grams and compares them against frequency models. These models are built from known legitimate domains (e.g., Alexa Top 1M) and

known DGA-generated domains. A domain is flagged as suspicious if its n-gram patterns match those commonly seen in malicious domains.

- **External Metadata (WHOIS):** Include information like a domain's creation date, registration length, and registrar details. These are often used to enrich the feature set. However, access to WHOIS data may become limited due to privacy regulations such as GDPR.

2.3. Detection Approach 2: Deep Learning Models

To overcome the limits of manual feature engineering, recent research has shifted toward deep learning. These models can automatically extract patterns from the raw character sequence of a domain name, without needing predefined features [2].

- **Sequential Models:** Since domain names are character sequences, Recurrent Neural Networks (RNNs) such as Long Short-Term Memory (LSTM) models are well-suited for DGA detection. Among these, Bidirectional LSTM (BiLSTM) models have shown better performance than standard LSTMs, as they process the domain string in both forward and backward directions, allowing the model to capture more contextual patterns [2].
- **Hybrid & Ensemble Models:** The literature shows that this often yields the best results. A common approach combines a Convolutional Neural Network (CNN) with a BiLSTM. The CNN captures local character patterns, while the BiLSTM learns the overall structure of the domain. Together, they provide both local and global context, improving detection accuracy. Another powerful technique is ensembling, where the outputs of different models (e.g., a deep learning model and a feature-based model like XGBoost) are combined in a final layer [5].

2.4. Detection Approach 3: Behavioral and Contextual Analysis

This aims to identify active **C2 communication** by analyzing the behavioral and contextual patterns of network traffic and host activity.

- **Graph-Based Analysis:** The GMAD framework introduced the Domain Name Travel Graph (DNTG), which maps the sequence of DNS queries made by clients. By analyzing sequential correlations between domains, it identifies

clusters that are frequently accessed together. This method is effective against stealthy, low-and-slow attacks, where malware spreads its activity over time to avoid detection [6].

- **Flow-Based Analysis:** This method analyzes statistical properties of DNS traffic grouped by a source IP over short time windows. Features like a very low DNS Success Ratio (many failed queries) are strong indicators of botnet activity [4].
- **Host-Based Analysis:** This approach links DNS activity to specific host-level events. It includes building detailed Program-DNS profiles for legitimate applications and detecting deviations from expected behavior. Some methods also track process-specific features, such as parent-child process relationships, to identify abnormal or malicious activity. This is one of the key features of supply chain attacks [7].

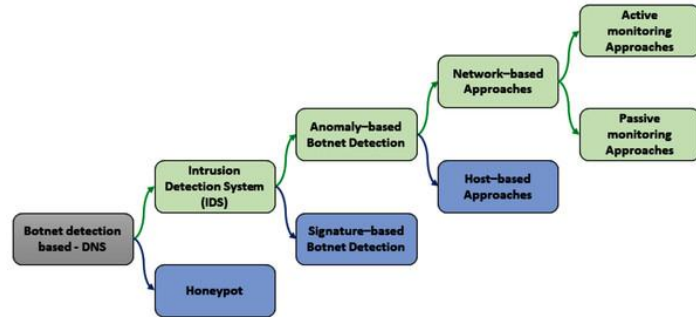


Figure 2: A Taxonomy of Detection Methods

3. RESEARCH GAP

A review of the literature reveals several key gaps in current DGA and C2 detection methodologies. These gaps include both high-level architectural divides between research approaches and specific limitations that authors have identified in their own state-of-the-art studies.

A key gap in existing detection systems is their isolated design. Most solutions in the literature are highly specialized, focusing either on the lexical structure of domain names or on behavioral patterns in network traffic [1]. Deep learning models are effective at classifying domain strings, while graph-based methods are effective at mapping C2 infrastructure [6]. However, these systems usually operate independently. This lack of integration misses a critical opportunity for synergy, where outputs from one method could enhance the precision and context of the other.

Another critical weakness lies in many behavioral detection techniques. Graph-based frameworks like GMAD are effective but often depend on a pre-existing blacklist to provide an initial "seed" for identifying malicious domain clusters. This reliance on known threats makes these systems reactive by design, reducing their ability to detect entirely new malware campaigns using previously unseen domains [6].

Finally, two major gaps are evident when considering real-world deployment. Many existing DNS analysis methods break down with the rise of encrypted protocols. Attackers now commonly hide C2 activity within standard TLS/SSL traffic, making it impossible for content-based detectors to inspect DNS queries [7]. This project addresses the detection of encrypted C2 communication as a distinct and critical problem within the broader DNS threat landscape.

A major gap also exists between high-performing academic models and their use in real-world operations. Research often focuses on accuracy metrics but overlooks how these models will integrate with Security Information and Event Management (SIEM) systems. As a result, many solutions fail to deliver context-rich, actionable alerts, something essential for security operations centers (SOCs) to detect and respond to threats effectively.

The following table compares our proposed framework against a selection of representative state-of-the-art research papers.

Research Paper	Synergistic Architecture (DGA-to-C2 Seeding)	Hybrid DGA Detection (Rules + Deep Learning)	C2 Graph-Based Analysis (for Stealthy Attacks)	Encrypted C2 Detection (via Metadata)	Operational SIEM/TI Integration
GMAD	Not Implemented	Not Implemented	Implemented	Not Implemented	Not Implemented
Efficient deep learning	Not Implemented	Implemented	Not Implemented	Not Implemented	Not Implemented

Hybrid rule-based	Not Implemented	Not Implemented	Not Implemented	Not Implemented	Not Implemented
C2-Eye	Not Implemented	Implemented	Not Implemented	Not Implemented	Not Implemented
Proposed Framework	Implemented	Implemented	Implemented	Implemented	Implemented

Table 1: DGA detection techniques comparison.

4. RESEARCH PROBLEM

Based on the gaps identified in existing literature, this research addresses a central, multi-layered problem that current solutions fail to fully resolve. The main challenge can be summarized by the following research question:

How can a synergistic, multi-component framework be designed to detect DGA and C2 threats by intelligently combining lexical and behavioral analysis, while also addressing the operational challenges of encrypted C2 communication and SIEM integration to produce useful, high-confidence alerts?

Answering this question requires combining multiple state-of-the-art techniques and focusing on the practical, real-world deployment of the final system.

5. OBJECTIVES

5.1. Main Objective

To design, implement, and evaluate a synergistic, multi-component framework for detecting DGA and C2 threats over DNS. The system will support encrypted C2 communication traffic metadata analysis and integrate with the Wazuh SIEM to deliver high-confidence, actionable alerts suitable for real-time operational use.

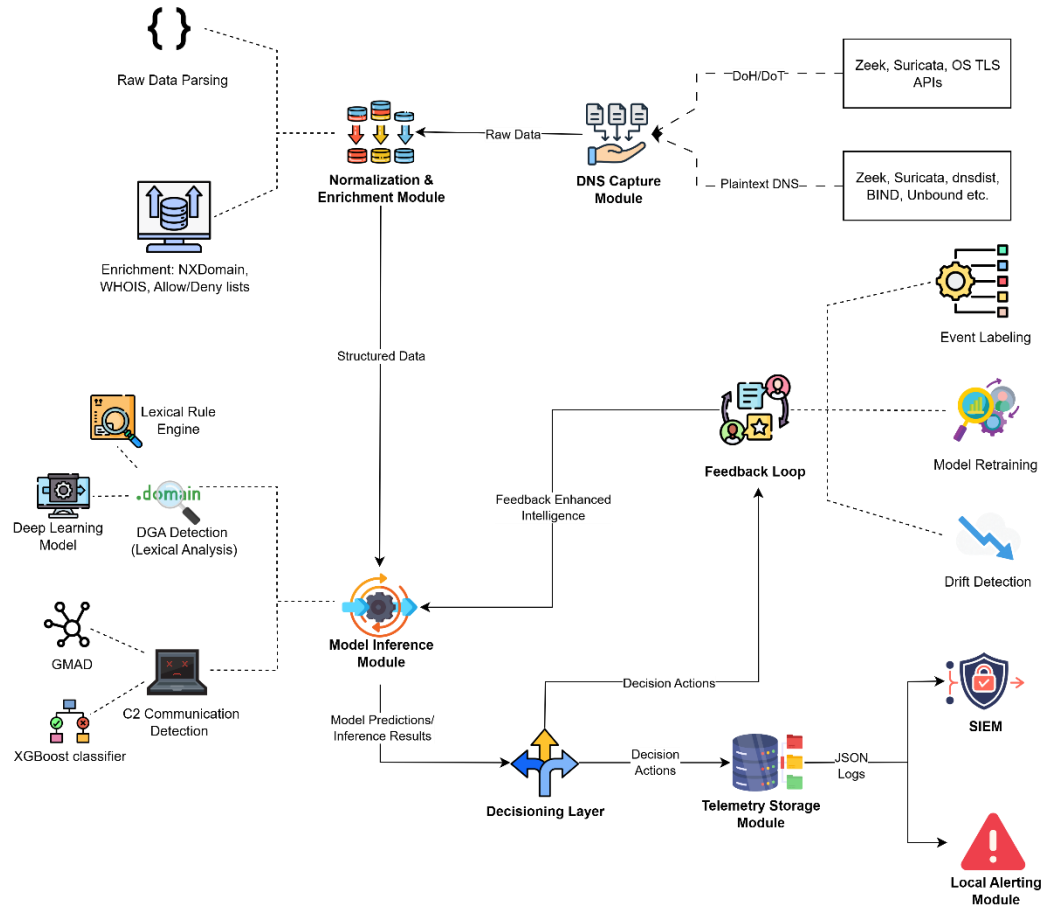
5.2. Specific Objectives

- To develop a cascaded DGA detection pipeline that integrates a transparent lexical rule engine (“Glass Box”) with a CNN+BiLSTM deep learning model (“Black Box”) to enable accurate, real-time classification of domain strings.
- To implement a C2 communication detection component that uses both graph-based infrastructure mapping to detect stealthy attacks and a flow-based XGBoost model to identify high-volume botnet anomalies in DNS traffic.
- To create a synergistic link between the DGA and C2 modules, where real-time DGA detections dynamically seed the graph-based C2 engine, addressing the "seed problem" faced by standalone graph models.
- To design and develop a specialized module for detecting encrypted C2 communication by analyzing unencrypted TLS metadata, addressing a key gap in current research.
- To integrate the framework's alerts with a Wazuh SIEM for centralized monitoring and visualization, and to build a feedback loop with a Threat Intelligence platform, demonstrating operational readiness.
- To evaluate the system using standard performance metrics, including Precision, Recall, and F1-Score, on benchmark datasets to validate its effectiveness compared to existing detection methods.

6. METHODOLOGY

This chapter details the complete technical approach for the project. It outlines the system architecture, the design of each component, the data collection strategy, and the project timeline required to achieve the research objectives.

6.1. System Architecture



6.2.Detailed Component Design & Tasks

This section outlines the specific implementation plan for each component of the framework, covering the core tasks and methodologies required to achieve the project's objectives.

6.2.1. Part 1: DGA Detection (Lexical Analysis)

The DGA detection component is designed as a two-tier "funnel" to analyze domain strings with both speed and accuracy.

Task 1: Implement the "Glass Box" Lexical Rule Engine

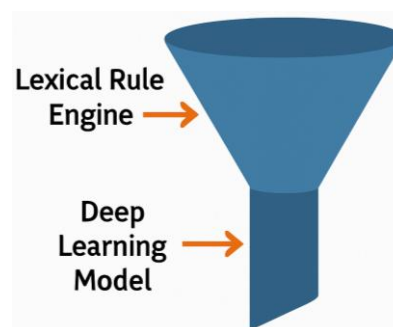
This task involves developing a custom, high-speed **"Glass Box" rule engine**. The engine will be designed to be similar in concept to professional detection rule languages like **YARA-L**, allowing for the creation of structured, high-confidence rules to act as a filter for obvious DGAs. The rules will be based on known DGA characteristics, including:

- **Shannon Entropy:** To measure the character randomness of the domain string, as high entropy is a strong indicator of algorithmic generation [4].
- **Anomalous Character Ratios:** To calculate the proportion of numbers to letters, which often deviates from normal linguistic patterns in DGA domains [1].
- **N-gram Scoring:** To compare a domain's substrings (n-grams) against frequency models of legitimate domains and known DGA domains. A domain that statistically resembles known bad domains will be flagged [8].

Task 2: Develop the "Black Box" Deep Learning Model

This model will analyze domains that pass through the rule engine to detect more complex and evasive DGAs.

- **Architecture:** The project will implement a CNN+BiLSTM hybrid model, a state-of-the-art architecture that has been shown to achieve the highest performance in DGA classification [2].
- **Implementation:** The model will use a character-level embedding layer to turn the domain string into a sequence of numbers. CNN layers will capture short character patterns, and a BiLSTM layer will then learn the overall structure of the domain. This bidirectional setup performs better than standard LSTM models.



6.2.2. Part 2: C2 Communication Detection (Behavioral Analysis)

This component focuses on the behavioral patterns of DNS traffic over time to identify C2 activity.

Task 3: Implement Graph-Based Infrastructure Mapping

To detect stealthy, "low-and-slow" attacks, this task involves implementing the **GMAD** methodology.

- **Methodology:** A Domain Name Travel Graph (DNTG) will be built using DNS logs, where each domain is a node and edges represent sequential queries made by the same clients. By identifying suspiciously linked domain sets in this graph, the system can reveal an attacker's full C2 infrastructure, even when the traffic is sparse or spread out over time [6].

Task 4: Implement Flow-Based Anomaly Detection

To detect noisier, high-volume botnet activity, this task involves building an XGBoost classifier.

- **Methodology:** DNS traffic will be aggregated into short time windows (e.g., 5 minutes) and grouped by the source IP address [4].
- **Features:** The XGBoost model will be trained on statistical features extracted from DNS flows. These include the DNS Success Ratio (low values often indicate bots querying unregistered domains), Average Domain Entropy, and Query Timing Entropy, which captures how regular or unpredictable the request timings are.

6.2.3. Part 3: Advanced Capabilities & SIEM Integration

This component addresses modern operational challenges to ensure the framework is practical and effective.

- **Methodology:** A rule-based classifier will be designed to flag suspicious connections based on:

- TLS Certificate Analysis: Checking for indicators like self-signed certificates or unusually short validity periods.
- TLS Fingerprinting (JA3/JA3S): Identifying non-standard client applications often used by malware.
- Connection Behavior: Analyzing the timing and data volume of sessions to detect the periodic, heartbeat-like beaconing characteristic of C2 communication.

Task 6: Implement SIEM & TI Integration

To make the findings operational, the framework will be integrated with external security tools.

- **SIEM Integration & Alerting:** All findings from the detection components will be formatted and forwarded to a Wazuh SIEM. A crucial part of this process is an intelligent alert prioritization system that assigns a severity level to each finding based on the confidence of the detection and the severity of the behavior. The levels will be Informational, Low, Medium, High, and Critical.
- **TI Integration:** A two-way API integration with a Threat Intelligence platform will be developed for enrichment (adding external context to internal alerts) and feedback (contributing newly discovered threats back to the security community).

6.3. Data Collection and Datasets

This project requires a diverse set of benign and malicious data to train and evaluate the different components of the framework. The data will be sourced from a combination of well-established public datasets.

6.3.1. Data for DGA Detection

The DGA detection component requires large lists of both legitimate and algorithmically generated domain names for training the lexical models.

- **Benign Domains:** To build a model of normal, human-generated domains, this project will use the Tranco Top 1 Million list or the Alexa Top 1M list. This is a standard and widely used academic practice for obtaining a baseline of legitimate web traffic.
- **Malicious DGA Domains:** To train the models to recognize malicious patterns, the project will use publicly available, labeled DGA domain datasets. A primary source will be the Netlab 360 DGA Data repository, which provides domains from a wide variety of known DGA families.

6.3.2. Data for C2 Communication Detection

The C2 detection component requires full network traffic logs, not just domain lists, to analyze behavioral patterns like query sequences and traffic flows.

- **Network Traffic Captures:** This project will use the public **CTU-13 dataset**. This well-known dataset contains real-world network traffic captures from 13 different botnet scenarios, providing the rich, contextual logs needed to train and validate the graph-based (GMAD) and flow-based (XGBoost) detection models.

6.3.3. Data for Encrypted C2 Analysis

Publicly available, labeled datasets for encrypted C2 traffic are rare. To address this, the project will generate a small, high-quality, custom dataset for developing and testing the encrypted C2 detection module.

- **Methodology:** A sandboxed virtual environment will be used to safely execute both common benign applications and known malware samples that communicate over TLS/SSL. The resulting encrypted traffic will be captured, and the unencrypted TLS metadata will be extracted and labeled to create a dataset for training the detection rules.

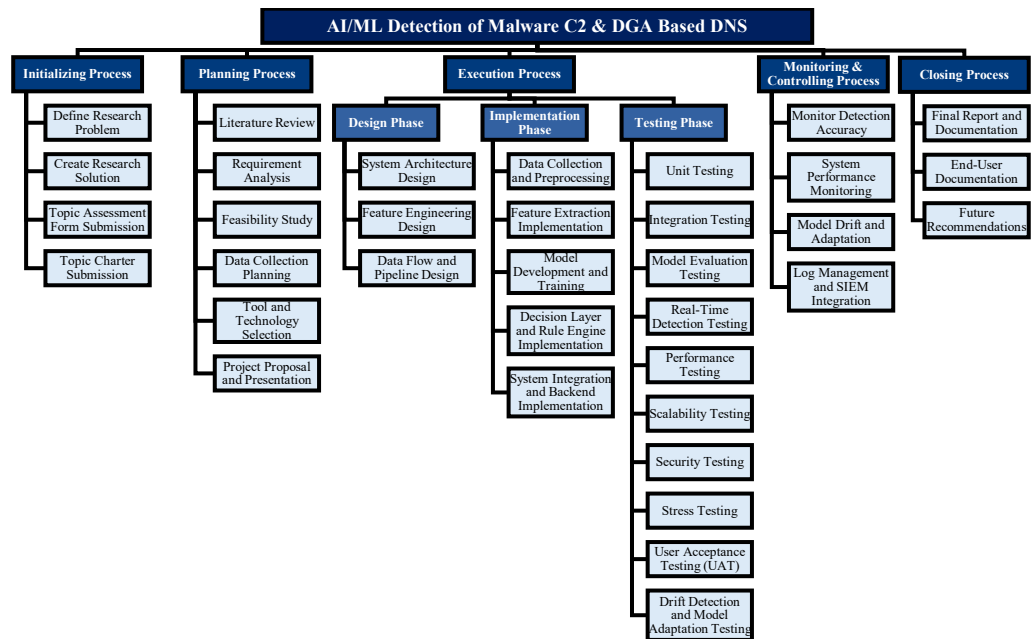
6.4. Tools and Technologies

This project will be developed using a combination of open-source software, established academic methodologies, and standard hardware.

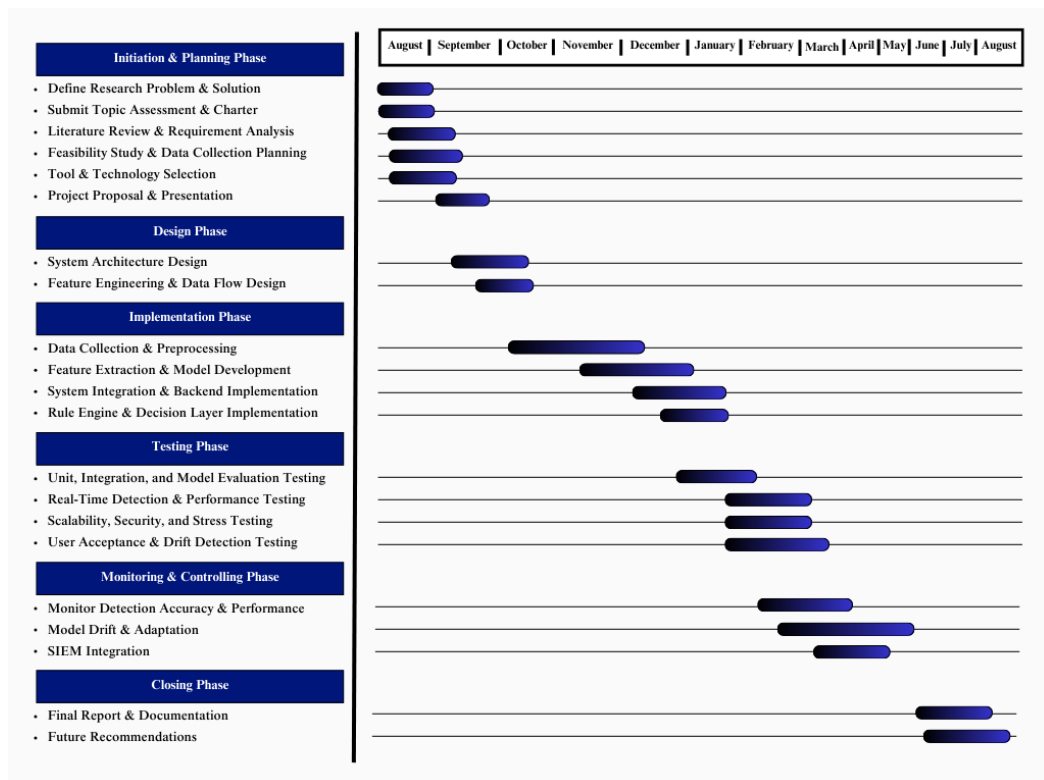
Technology	Purpose
Python	The core programming language for all custom development, data processing, and model implementation.
Pandas & Scikit-learn	Core libraries used for data manipulation, feature engineering from DNS logs, and supporting machine learning tasks.
TensorFlow with Keras	The deep learning framework for building, training, and evaluating the CNN+BiLSTM model [1].
XGBoost	The high-performance library used to build the classifier for the flow-based C2 anomaly detection component [5].
NetworkX	The Python library used to construct and analyze the graph structures required for the GMAD methodology.
Wazuh (ELK Stack)	The open-source SIEM platform, built on the ELK Stack, that will be used for alert ingestion, correlation, and creating visualization dashboards.
Hardware & Environment	The project will be developed on a personal computer. A suitable cloud, like Google Colab, will be used to access GPU resources for the efficient training of the deep learning model.

6.5. Project Timeline

6.5.1. Work Breakdown Structure (WBS)



6.5.2. Gantt Chart



6.6. Anticipated Conclusion

The research is anticipated to conclude with the following key deliverables and validated outcomes:

- **A Functional Prototype:** The project will deliver a working prototype of the entire detection framework, demonstrating the successful integration of all lexical and behavioral components with the Wazuh SIEM.
- **Benchmark-Proven Superiority:** Final evaluation is expected to show that the proposed synergistic architecture produces more accurate, context-aware, and high-confidence alerts compared to standalone or specialized detection methods commonly found in existing research.
- **An Operational Blueprint:** A practical blueprint for a real-world security tool, demonstrated by its successful integration with a Wazuh SIEM to provide actionable intelligence for security analysts.
- **A Focus on High Detection Efficacy (High Recall):** The evaluation will prioritize achieving a high **Recall** rate, as minimizing False Negatives (missed threats) is critical for security applications. This will validate the framework's efficacy in catching the vast majority of malicious domains, a goal of great importance in DGA detection.

7. PROJECT REQUIREMENTS

7.1. Functional Requirements

- **Data Ingestion:** The system shall ingest DNS query logs and network traffic data from standard sources (e.g., .pcap files, log files).
- **Lexical Rule-Based Detection:** The system shall apply a set of pre-defined lexical rules (e.g., entropy, n-gram scores) to classify domain strings.
- **Deep Learning Detection:** The system shall classify domain strings using a trained CNN+BiLSTM model to identify complex DGAs.

- **Graph-Based C2 Detection:** The system shall construct a Domain Name Travel Graph (DNTG) from DNS query sequences to identify clusters of malicious domains indicative of C2 infrastructure.
- **Encrypted C2 Detection:** The system shall analyze TLS session metadata to identify indicators of C2 communication within encrypted traffic.
- **Alert Prioritization:** The system shall generate prioritized alerts with severity levels (Informational to Critical) based on correlated findings from its different components.
- **SIEM Integration:** The system shall forward all generated alerts in a structured format (e.g., JSON) to a Wazuh SIEM for visualization and further correlation.
- **Threat Intelligence Integration:** The system shall be able to enrich alerts by querying an external Threat Intelligence platform with identified indicators.

7.2. Non-Functional Requirements

- **Performance:** The DGA detection components shall operate in near real-time. The behavioral C2 components shall process log data on time, suitable for operational analysis.
- **Accuracy:** The classification models shall achieve a high F1-Score when evaluated on benchmark datasets.
- **High Recall:** The overall framework shall be optimized for a high **Recall** rate to minimize False Negatives (missed threats), which is a critical goal for DGA detection systems.
- **Usability:** The final alerts, when viewed in the Wazuh dashboard, shall be structured and context-rich, providing actionable information for a security analyst.

7.3. User Requirements

- **Prioritized Threat View:** The user needs to view alerts prioritized by severity (e.g., Critical, High, Medium) in the SIEM dashboard to focus on the most urgent threats first.
- **Access to Contextual Alert Data:** The user must be able to investigate any alert to see the full context.

- **Situational Awareness Dashboards:** The user needs to visualize security trends on the Wazuh dashboard, including charts for top DGA detections and timelines of C2 activity.

7.4. System Requirements

Define the necessary hardware and software environment needed to run the detection framework.

7.4.1. Hardware Requirements

Component	Specification
Development & Hosting Machine	A personal computer with a modern multi-core CPU (e.g., Intel Core i7 or equivalent) and a minimum of 16GB of RAM.
GPU Access	Required for deep learning model training, to be facilitated through a cloud service (e.g., Google Colab).

7.4.2. Software Requirements

Category	Software / Library
Operating System	64-bit OS (e.g., Ubuntu 20.04+ or Windows 10+).
Core Platforms	Python 3.8+, Wazuh SIEM (Manager and Dashboard).
Key Python Libraries	TensorFlow, Keras, XGBoost, Pandas, Scikit-learn, NetworkX.

7.5. Use Cases (Tentative)

Use Case 1: Investigate a High-Priority DGA Alert

- **Actor:** Security Analyst
- **Goal:** To investigate a high-severity alert to understand its full context and determine the required response.

- **High-Level Steps:**

1. The Analyst observes a "High" or "Critical" DGA-related alert in the Wazuh dashboard.
2. The Analyst opens the alert to view the correlated details provided by the framework.
3. The system displays the malicious domain, the reasons for the flag (e.g., "Matched High Entropy Rule," "CNN+BiLSTM Score: 0.99"), the internal host that made the query, and any Threat Intelligence enrichment.
4. The Analyst uses this complete picture to initiate an incident response process, such as isolating the affected host.

Use Case 2: Proactively Hunt for C2 Infrastructure

- **Actor:** Security Analyst

- **Goal:** To use the framework's behavioral analysis tools to identify a potential "low-and-slow" C2 campaign.

- **High-Level Steps:**

1. The Analyst navigates to a custom "Threat Campaigns" dashboard in Wazuh.
2. The system displays visual graphs of DNS clusters identified by the GMAD component.
3. The Analyst selects a cluster of interest for further investigation.
4. The system displays the full graph of related C2 domains, the timeline of activity, and a list of all internal hosts that have communicated with the cluster.
5. The Analyst uses this intelligence to identify the stealthy attack and block the entire C2 infrastructure.

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